cloudtorch / develop — Step 5a: the velocity collapse is the regularisation

2026-08-24 — controlled 2×2 factorial isolating why 4a's velocity maps have structure and 5a's collapse. **Corrects** `step_5a_debug.pdf`: the earlier note concluded "the real lever is *not* a reg knob." That was wrong — the explicit regularisation is exactly the lever.

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1. WHAT WAS RUN

A clean 2×2 on **fixed frames** [11–14] (4a's window), the **committed** `pinn_model.py` init ($v_{\rm los} = \mp 20$, scale 20 — what produced the stored 4a/5a outputs, restored by monkeypatch), and every other knob held constant (`N_FREQ`=256, `SIGMA_T`=4, `WIDTH`=256, `DEPTH`=4, `S_MAX`=0.7, `DV_FLOOR`=8, `LR`=3e-3, 3000 iters, seed 0). **Only two things vary:**

- `SIGMA_XY` $\in \{12, 32\}$ — the Fourier spatial bandwidth (implicit smoothness);
- regularisation $\in$ {off (4a), on (5a's original `REG`, `w_xy`=1e-3)}.

Runs: A=(32, off) ≈ 4a, B=(12, off), C=(32, on), D=(12, on) ≈ 5a. Script: `scratchpad/ab_4a_vs_5a.py`.

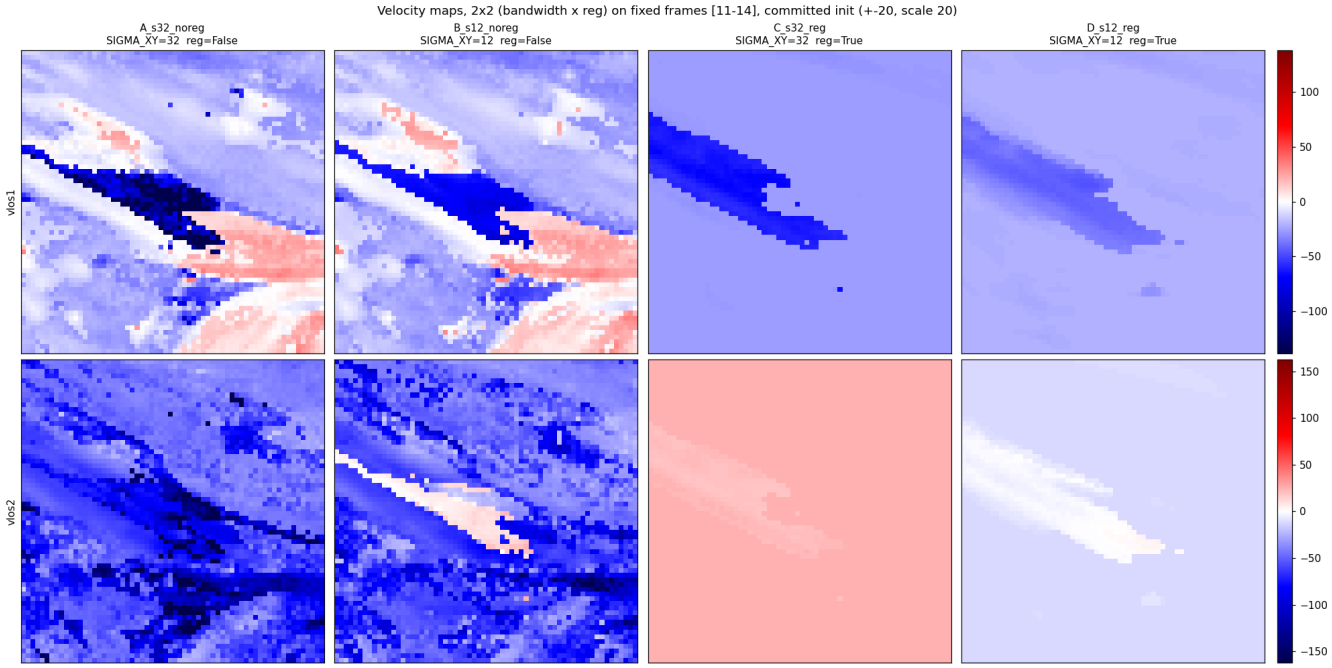
2. RESULT

run	`SIGMA_XY` / reg	$\langle\chi^2\rangle$	RMS	v_1 [min/med/max]	v_2 [min/med/max]
A	32 / **off**	0.252	3.96%	$-226/-17/+\mathbf{52}$	$-308/-56/-10$
B	12 / **off**	0.195	3.48%	$-96/-16/+\mathbf{51}$	$-183/-54/+\mathbf{24}$
C	32 / **on**	0.604	6.13%	$-71/-27/-\mathbf{27}$	$+18/+25/+\mathbf{25}$
D	12 / **on**	0.904	7.50%	$-47/-21/-\mathbf{21}$	$-13/-12/+4$

Read the last two columns: reg **off** ⇒ velocities span a wide, *sign-crossing* range (structure); reg **on** ⇒ min ≈ med ≈ max (a single collapsed value). This holds at *both* bandwidths.

3. FINDING: REG DECIDES COLLAPSE; BANDWIDTH DOES NOT

Toggling the regularisation flips structured ↔ collapsed at either `SIGMA_XY`; changing `SIGMA_XY` at fixed reg does *not*. So `SIGMA_XY` only sets how rough the maps get (higher = wilder tails: v_1 reaches -226 in A vs -96 in B); the **explicit regularisation is what collapses the velocity field**. As a bonus the reg also *worsens* the data fit here (χ^2 0.2 → 0.6–0.9).



$v_{\rm los1}$ (top) and $v_{\rm los2}$ (bottom) for the four runs, common colour scale per row. Reg-off (A, B): filamentary, sign-crossing flows. Reg-on (C, D): near-uniform.

4. WHY A ∼1–2% PENALTY DOMINATES (AND WHY THE EARLIER FIX MISSED)

The reg is only **0.8–2.0%** of the loss yet flattens the velocity completely, because $v_{\rm los}$ is a *near-flat, weakly-constrained* direction: the given background owns the line wings where the Doppler signal lives, so the data barely pulls on velocity. A tiny smoothness/hinge penalty then dominates that null direction and pulls the whole field to a constant. "Non-binding in loss magnitude" ≠ "harmless." This also explains why relaxing **only** `w_xy` 100× (the `step_5a_debug` test) didn't help: it left the *other* velocity terms — temporal `w_t`=2e-3 and the `vlos` range hinge — active, and *those* drive the collapse.

5. NEXT LEVERS

- **Cut the velocity reg as a whole** (`w_t` + hinge, not `w_xy` alone), ∼10–100× weaker, until structure survives while χ^2 stays low.
- **Sub-factorial** (`w_t`-only / hinge-only / `w_xy`-only) to pin down which term dominates — not yet isolated.
- Reg-off carries unphysical tails ($v_1 \to -226$, `dv1`→151): the aim is a penalty that *clips outliers without flattening* (robust loss or a far-outer hinge), not the current over-scaled velocity smoothing.

Runs used the *committed* `pinn_model.py` (∓ 20, scale 20). The working tree now carries an uncommitted asymmetric init ($-50/0$, scale 5), so neither notebook's stored output reproduces as-is. Figure: `docs/step_5a_factorial_vlos.png`.